



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

BIOLOGY

6030/2

PAPER 2 Theory Structured

JUNE 2023 SESSION

1 hour 30 minutes

Additional materials:
Electronic calculator

TIME: 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
TOTAL	

This question paper consists of 11 printed pages and 1 blank page.

Copyright: Zimbabwe School Examinations Council, J2023.



Answer all questions

1

Fig. 1.1 shows the structure of a membrane.

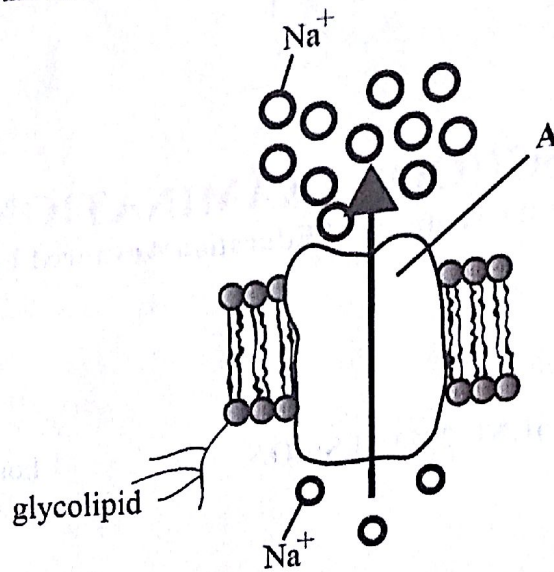


Fig. 1.1

- (a) (i) Describe how the part labelled A interacts with the phospholipids surrounding it.

[2]

- (ii) Name the other molecule required to complete the transport of Na^+ .

[1]

- (iii) State the reason why Na^+ ions **cannot** pass through the phospholipid bilayer.

[1]

- (b) Describe any **two** roles of glycolipids in the membrane.

[2]

[Total: 6]

(a) Fig. 2.1 shows the molecular structure of maltose.

For
Examiner's
Use

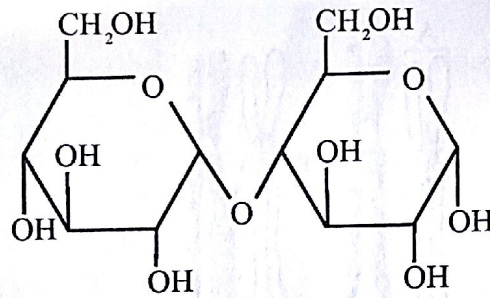


Fig. 2.1

- (i) Name the residue of maltose after breakdown.

[1]

- (ii) Draw a diagram to show the breakdown of maltose.

[2]

- (b) Explain how starch is adapted for its role as a storage material.

[3]

[Total: 6]



3

(a) Fig. 3.1 shows chromosome behaviour during nuclear division.

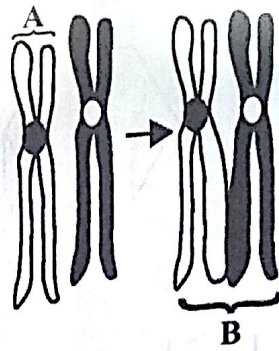


Fig. 3.1

(i) Identify structure A.

[1]

(ii) Describe the process occurring at B.

[2]

(iii) State the site of the type of nuclear division shown in Fig. 3.1 in humans.

[1]

(b) (i) State any **one** similarity between homologous chromosomes.

[1]

(ii) Explain the significance of mitosis in tissue repair.

[2]

[Total: 7]



- (a) (i) Name the site of transcription.

[1]

- (ii) State the role of mRNA and ribosomes during translation.

mRNA

ribosomes

[2]

- (b) Describe the process of translation.

[4]

[Total:7]

Fig.5.1 shows a reaction that occurs in red blood cells.

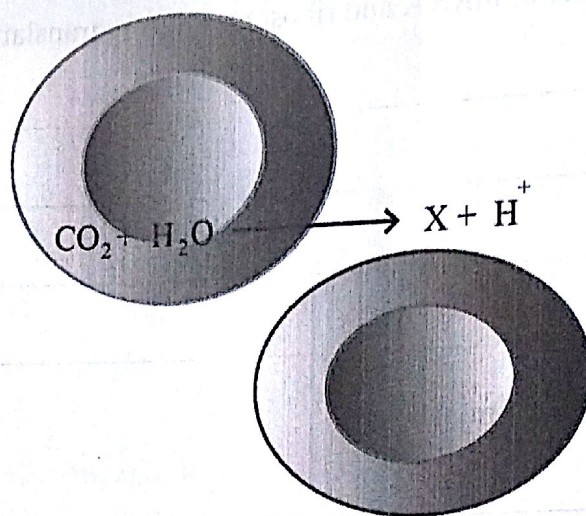


Fig. 5.1

- (a) (i) Identify molecule X.

_____ [1]

- (ii) Name the enzyme that catalyses the reaction.

_____ [1]

- (b) Describe the role of hydrogen ions (H^+) in oxygen carriage.

[3]

[Total: 5]

- 6 (a) Fig. 6.1 is a flow chart showing some stages of photosynthesis.

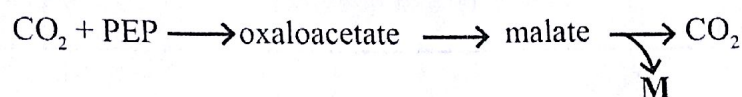


Fig. 6.1

- (i) State the enzyme that catalyses the reaction between CO_2 and PEP.

[1]

- (ii) Name the compound M.

[1]

- (b) Explain the significance of malate in C_4 photosynthesis.

[3]

[Total:5]

Observations made by Charles Darwin in the Theory of natural selection.

observation 1

observation 2

observation 3

[3]

(b) Give any **three** differences between natural and artificial selection.

1.

2.

3.

[3]

[Total:6]



- (a) Fig. 8.1 shows the cross section of part of an anther.

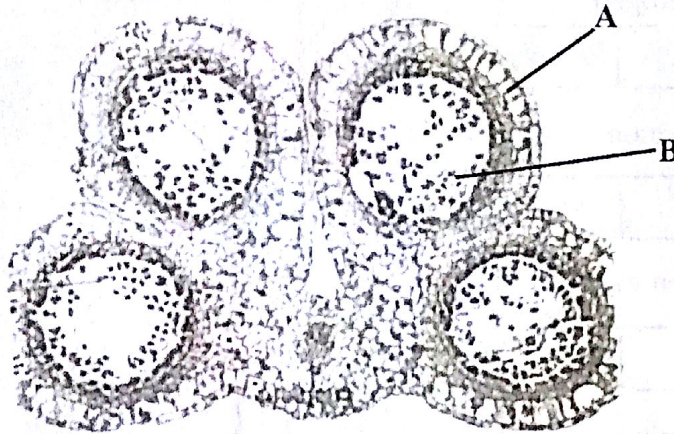


Fig. 8.1

- (i) Name the part labelled:

A _____ [1]

B _____ [1]

- (ii) State the role of part A.

_____ [1]

- (b) Describe how pollen grains are formed.

_____ [4]
[Total:7]



9 Fig. 9.1 shows part of the nitrogen cycle.

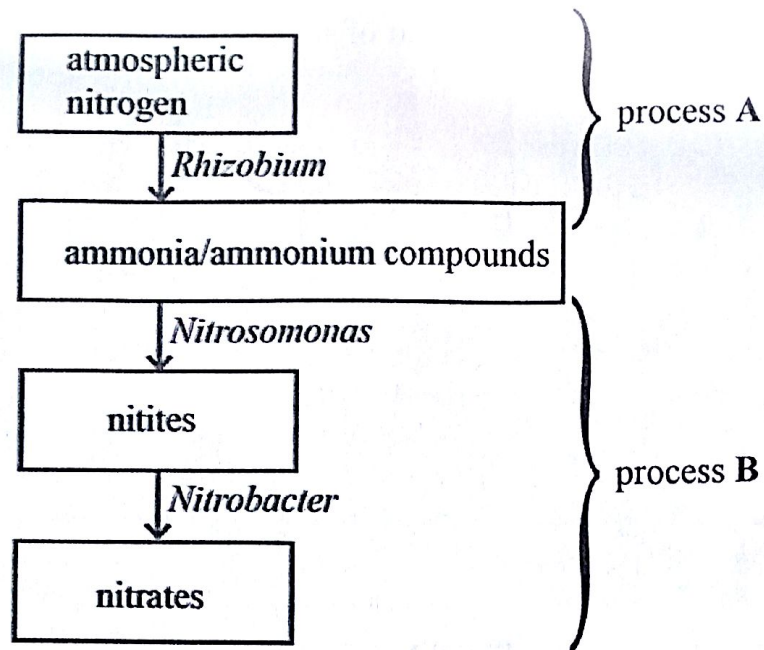


Fig. 9.1

(a) Identify process:

A _____

B _____

[2]

(b) Explain how the bacteria may be used in place of artificial fertilisers.

Fig. 10.1 shows a single celled organism.

For
Examiner's
Use

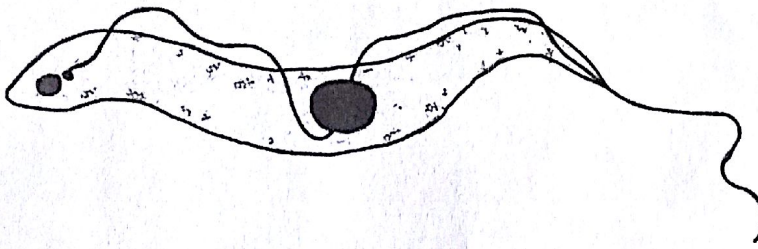


Fig. 10.1

(a) State any **three** visible characteristics of the organism.

1 _____

2 _____

3 _____

[3]

(b) Name the kingdom to which the organism belongs.

[1]

(c) State any **one** importance of organisms in the kingdom named in (b).

[1]

[Total:5]